

B.Sc. (Hons.) in Environmental Science

- B.Sc. (Hons.) in Environmental Science is an undergraduate program that focuses on the study of the natural environment and the impact of human activities on it.
- In this course, students learn to investigate the interaction of biological and physical systems in the environment.
- Some of the major topics students may study in this course are animal studies, climate, energy, water, soil, dynamics of the plants, local plants, animal populations and their communities.
- B.Sc. (Hons.) Environmental Science brings together knowledge from a variety of fields to give students a deep understanding of the environmental challenges facing the world today.
- Through a practical, hands-on approach that runs through all the modules, students gain exposure to the skills and technology employed at the forefront of environmental science.
- This degree explores the multiple disciplines needed to understand, manage and protect our planet. Students assess environmental problems, propose solutions, and gain a comprehensive understanding of the natural environment.
- The program covers a wide range of topics, including ecology, geology, hydrology, atmospheric science, conservation biology, and environmental management.

Unit 1 - Environment

Definition

The environment is the sum total of all the physical, chemical, and biological factors and conditions that determine the growth and survival of an organism or a community of organisms.

Types of Environment

There are two main types of environment: natural and man-made. Natural environments include forests, oceans, mountains, and other ecosystems that are not significantly altered by human activity. Man-made environments, on the other hand, are created by humans and include cities, towns, and other built-up areas.

Components of Environment

The environment is made up of various components, including: - **Abiotic components:** These are the non-living components of the environment, such as air, water, and soil. - **Biotic components:** These are the living components of the environment, such as plants, animals, and microorganisms.

Segments of Environment

The environment can be divided into various segments, including: - **Atmosphere**: This is the layer of gases that surrounds the Earth. - **Hydrosphere**: This is the total amount of water on the Earth's surface, including oceans, lakes, rivers, and groundwater. - **Lithosphere**: This is the solid outer layer of the Earth, including the crust and uppermost mantle. - **Biosphere**: This is the part of the Earth where life exists, including all living organisms and their interactions with the abiotic components of the environment.

Scope and Importance

The environment is important because it provides the conditions necessary for the survival of all living organisms. It also plays a crucial role in regulating the Earth's climate and weather patterns. Understanding the environment and its components is essential for making informed decisions about how to manage and protect it.

Need for Public Awareness

Public awareness about the environment is important because it helps individuals understand the impact of their actions on the environment and encourages them to make more sustainable choices. Public awareness campaigns can also help to promote environmental protection and conservation efforts.

Ecosystem

Definition

An ecosystem is a community of living and non-living things interacting with each other in a specific environment.

Types of Ecosystem

There are many different types of ecosystems, including: - **Terrestrial ecosystems**: These are ecosystems found on land, such as forests, grasslands, and deserts. - **Aquatic ecosystems**: These are ecosystems found in water, such as oceans, lakes, and rivers. - **Artificial ecosystems**: These are man-made ecosystems, such as urban parks and gardens.

Structure of Ecosystem

An ecosystem is made up of various components, including: - **Producers**: These are organisms that produce their own food through photosynthesis, such as plants. - **Consumers**: These are organisms that consume other organisms for food, such as animals. - **Decomposers**: These are organisms that break down dead organic matter, such as bacteria and fungi.

Food Chain

A food chain is a sequence of organisms in which each organism is eaten by the next one in the chain. For example, in a grassland ecosystem, the food chain might be: grass -> rabbit -> fox.

Food Web

A food web is a more complex representation of the feeding relationships within an ecosystem. It shows the multiple pathways through which energy and nutrients flow between different organisms.

Ecological Pyramid

An ecological pyramid is a graphical representation of the trophic levels (feeding levels) of an ecosystem. The base of the pyramid represents the primary producers, while the higher levels represent the consumers. The size of each level represents the relative amount of biomass or energy at that level.

Balance Ecosystem

A balanced ecosystem is one in which the populations of different species are in balance with each other and with the abiotic components of the environment. In a balanced ecosystem, the populations of predators and prey are stable, and the ecosystem is able to support a diverse range of species.

Effects of Human Activities on Environment

Human activities, such as food production, shelter, housing, agriculture, industry, mining, transportation, economic and social security, can have significant impacts on the environment. These impacts can include: - **Habitat destruction:** Human activities, such as deforestation and urbanization, can destroy natural habitats and reduce biodiversity. - **Pollution:** Human activities, such as burning fossil fuels and using pesticides, can release pollutants into the air, water, and soil, harming the environment and human health. - **Climate change:** Human activities, such as burning fossil fuels and deforestation, can release greenhouse gases into the atmosphere, contributing to global warming and climate change.

Environmental Impact Assessment

Environmental Impact Assessment (EIA) is a process used to evaluate the potential environmental impacts of a proposed project or development. The EIA process involves identifying and assessing the potential impacts of

Unit 2 - Natural Resources: Introduction, Classification

Natural resources are materials and substances that occur naturally within the environment and can be used for economic gain. They are classified into two main categories: renewable and non-renewable resources.

Renewable resources are those that can be replenished naturally in a relatively short period of time, such as sunlight, wind, and water. Non-renewable resources, on the other hand, are those that are finite and cannot be replenished, such as fossil fuels and minerals.

Water Resources

Water is one of the most important natural resources, as it is essential for life and is used in a wide variety of ways, including for drinking, agriculture, and industry. Water resources can be classified into two main categories: surface water and groundwater.

Surface water is found in rivers, lakes, and reservoirs, while groundwater is found in underground aquifers. The availability and quality of water resources can vary greatly depending on factors such as location, climate, and human activity.

Waterborne diseases are illnesses caused by microorganisms that are transmitted through contaminated water. These can include cholera, typhoid, and dysentery. Water-induced diseases, on the other hand, are illnesses that are caused by the lack of clean water, such as dehydration and malnutrition.

Fluoride and arsenic are two common contaminants found in drinking water. High levels of fluoride can cause dental and skeletal fluorosis, while high levels of arsenic can cause skin lesions and increase the risk of cancer.

Mineral Resources

Minerals are naturally occurring substances that are extracted from the earth for their economic value. They can be classified into two main categories: metallic and non-metallic minerals.

Metallic minerals, such as gold, silver, and copper, are typically extracted from ore deposits and are used in a wide variety of industries, including electronics, construction, and transportation. Non-metallic minerals, on the other hand, include substances such as sand, gravel, and clay, and are used in industries such as construction and agriculture.

Material Cycles

Material cycles refer to the natural processes by which elements and compounds are cycled through the environment. These cycles play a crucial role in maintaining the balance of the ecosystem and include the carbon, nitrogen, and sulfur cycles.

The carbon cycle involves the movement of carbon between the atmosphere, oceans, and land. Carbon dioxide is taken up by plants during photosynthesis and is released back into the atmosphere through respiration and decomposition.

The nitrogen cycle involves the movement of nitrogen between the atmosphere, soil, and living organisms. Nitrogen is converted into various forms, including ammonia, nitrate, and nitrogen gas, through processes such as nitrogen fixation, nitrification, and denitrification.

The sulfur cycle involves the movement of sulfur between the atmosphere, soil, and living organisms. Sulfur is released into the atmosphere through volcanic eruptions and the burning of fossil fuels and is taken up by plants and incorporated into organic matter.

Energy Resources

Energy resources are sources of energy that can be used to generate electricity, heat, and motion. They can be classified into two main categories: conventional and non-conventional sources of energy.

Conventional sources of energy include fossil fuels, such as coal, oil, and natural gas, and nuclear energy. These sources of energy are non-renewable and have a significant impact on the environment.

Non-conventional sources of energy, on the other hand, include renewable sources such as solar, wind, and hydro power. These sources of energy have a much lower impact on the environment and are becoming increasingly popular as the world moves towards a more sustainable future.

Forest Resources

Forests are an important natural resource, providing a wide range of benefits, including timber, non-timber forest products, and ecosystem services. However, forests are under threat from deforestation and degradation, which can have significant impacts on the environment and society.

Deforestation refers to the permanent removal of forests, while forest degradation refers to the reduction in the quality of forests. These processes can have a range of negative impacts, including the loss of biodiversity, increased greenhouse gas emissions, and reduced water quality.

It is important to manage forest resources sustainably in order to ensure their continued availability and to minimize their impact on the environment. This can be achieved through measures such as sustainable forest management, reforestation, and the protection of forested areas.

Unit 3 - Pollution and their Effects; Public Health Aspects of Environmental; Water Pollution, Air Pollution, Soil Pollution, Noise Pollution, Solid waste management.

Pollution is the introduction of harmful substances or products into the environment. It is a major problem in America and as well as the world. Pollution not only damages the environment, but it also damages the health of human beings.

1. **Water Pollution:** Water pollution is the contamination of water bodies such as lakes, rivers, oceans, and groundwater. This form of environmental degradation occurs when pollutants are directly or indirectly discharged into water bodies without adequate treatment to remove harmful compounds. Water pollution affects the entire biosphere – plants and organisms living in these bodies of water.
2. **Air Pollution:** Air pollution is the introduction into the atmosphere of chemicals, particulates, or biological materials that cause harm or discomfort to humans or other living organisms, or damages the natural environment. Air pollution can result from both human and natural actions.
3. **Soil Pollution:** Soil pollution is the presence of man-made chemicals or other alteration in the natural soil environment. This type of contamination typically arises from the rupture of underground storage tanks, application of pesticides, percolation of contaminated surface water to subsurface strata, oil and fuel dumping, leaching of wastes from landfills or direct discharge of industrial wastes to the soil.
4. **Noise Pollution:** Noise pollution is excessive, displeasing human, animal, or machine-created environmental noise that disrupts the activity or balance of human or animal life. The word noise comes from the Latin word nausea, meaning seasickness.
5. **Solid Waste Management:** Solid waste management is the collection, transport, processing, recycling or disposal, and monitoring of waste materials. The term usually relates to materials produced by human activity, and is generally undertaken to reduce their effect on health, the environment, or aesthetics.

These are the main types of pollution and their effects on public health and the environment. It is important to take steps to reduce pollution and protect our environment and health.

Unit 4 - Current Environmental Issues of Importance

1. **Global Warming:** This is an increase in the ambient temperature of Earth which will negatively affect life on Earth. The excessive burning of fossil fuels such as petrol, coal, etc. have resulted in an increase in the

number of greenhouse gases in the atmosphere resulting in a phenomenon known as Global Warming.

2. **Green House Effects:** The greenhouse effect happens when certain gases—known as greenhouse gases —collect in Earth’s atmosphere. The greenhouse gases responsible for the greenhouse effect are: Water Vapour, Carbon Dioxide, Methane, Ozone.
3. **Climate Change:** Human activities contribute to global warming by increasing the greenhouse effect. Even slight rises in average global temperatures can have huge effects. Perhaps the biggest, most obvious effect is that glaciers and ice caps melt faster than usual. The meltwater drains into the oceans, causing sea levels to rise and oceans to become less salty.
4. **Acid Rain:** The effects of acid rain, combined with other environmental stressors, leave trees and plants less healthy, more vulnerable to cold temperatures, insects, and disease.
5. **Ozone Layer Formation and Depletion:** Greenhouse gases include carbon dioxide, methane, nitrous oxides, and water vapor. Scientists have determined that carbon dioxide’s warming effect helps stabilize Earth’s atmosphere.
6. **Population Growth and Automobile pollution:** N/A
7. **Burning of paddy straw:** N/A

Unit 5 - Environmental Protection

Environmental Protection Act 1986

- The Environmental Protection Act of 1986 is an act of the Parliament of India.
- It was enacted to provide for the protection and improvement of the environment and for matters connected therewith.
- The act is an umbrella legislation designed to provide a framework for central government coordination of the activities of various central and state authorities established under previous laws, such as the Water Act and the Air Act.

Initiatives by Non Governmental Organizations (NGO’s)

- Non-governmental organizations (NGOs) play a vital role in the protection and conservation of the environment.
- NGOs work at the grassroots level to create awareness and mobilize local communities to take action to protect the environment.
- Some of the initiatives taken by NGOs include tree planting, waste management, and conservation of natural resources.

Human Population and the Environment

Population growth

- The rapid growth of the human population has had a significant impact on the environment.
- Population growth leads to increased demand for resources, such as food, water, and energy, and increased production of waste and pollution.
- This can result in habitat destruction, deforestation, and loss of biodiversity.

Environmental Education

- Environmental education is the process of teaching individuals about the environment and how to protect it.
- It aims to increase awareness and understanding of environmental issues and to encourage individuals to take action to protect the environment.
- Environmental education can take place in formal and informal settings, such as schools, community centers, and nature reserves.

Women Education

- Educating women is crucial for the protection of the environment.
- Women play a vital role in managing natural resources and promoting sustainable development.
- Educating women can help to empower them to take action to protect the environment and to promote sustainable development.